NeoCraft Macro Desk — User Manual

A custom macro pad: a 4×4 button grid plus 3 rotary encoders, paired with a Windows desktop app that decides what each button and encoder does.

Part 1 — What it does

The hardware

- **16 buttons** arranged in a 4×4 grid. 15 of them (`BTN0` – `BTN14`) run actions you assign. The 16th, `2FX`, is not an action button — it's a layer switch (see below).
- **3 rotary encoders** (`ENC1` – `ENC3`). Turning one adjusts a volume; clicking it in mutes/unmutes. *Currently only ENC1 is wired up in the firmware — ENC2 and ENC3 are physically present but not active yet.*
- **Addressable LEDs**, one per key, showing an animated color pattern while everything's connected, or **solid red** if the app isn't running/connected — a built-in reminder to start the app.

The app

A small program that lives in your system tray. It talks to the pad over USB and lets you assign an action to every button, and a volume target to every encoder. Each button can do one of these:

Action type	What it does
Keyboard	Sends a key combo (e.g. `ctrl+c`) to whatever app is focused
Macro	Same as Keyboard, but you record the combo by pressing it live instead of typing it
OBS Scene	Switches OBS Studio to a specific scene (requires OBS running with its WebSocket server on)
Sound	Plays an audio file (WAV, MP3, OGG, FLAC, AIFF) — with volume and trim controls
Empty	Does nothing (the button's default state)

Each encoder can control either your **system's overall volume**, or the volume of **one specific application** (even multi-window apps like a browser) — your choice, set per encoder.

The "2FX" second layer

Every button can hold **two different actions** — a normal one (Layer 1) and a second one (Layer 2) that only fires when you deliberately arm it:

1. Tap `2FX` — the border flashes red, Layer 2 is now armed.
2. Tap any other button — it runs *that button's Layer 2 action*, then the pad automatically drops back to Layer 1.
3. If you don't press anything, Layer 2 auto-cancels after a timeout (10 seconds by default, adjustable up to 60s).
4. Tapping `2FX` again while armed cancels it manually, with no action run.

This is how one 16-key pad gives you effectively 30 assignable actions.

Background operation

Closing the app window (the **X** button) doesn't quit it — it minimizes to the system tray and keeps running, so your button/encoder mappings stay active. Use the tray icon's **Quit** option to fully exit.

Language

The app is available in **English** and **Português**, switchable at any time from Settings (see Changing the app language). Defaults to Português. *Note: the screenshots in this manual were captured before this feature existed, so a few labels shown (dialog titles, button text) are in whichever language they defaulted to at the time — the dialog layout and flow are identical, only the wording changes per language now.*

Part 2 — Installation

1. Download `NeoCraft-Macro-Desk-Setup.exe`.
2. Double-click it. **No administrator prompt will appear** — this installs just for your Windows account, not system-wide.
3. Optionally check **"Create a desktop shortcut"** in the wizard.
4. Click through to finish. The installer creates a Start Menu entry (**NeoCraft Macro Desk**) and an uninstaller.
5. Plug in the macro pad via USB.
6. Launch the app (Start Menu, or the desktop shortcut if you created one).

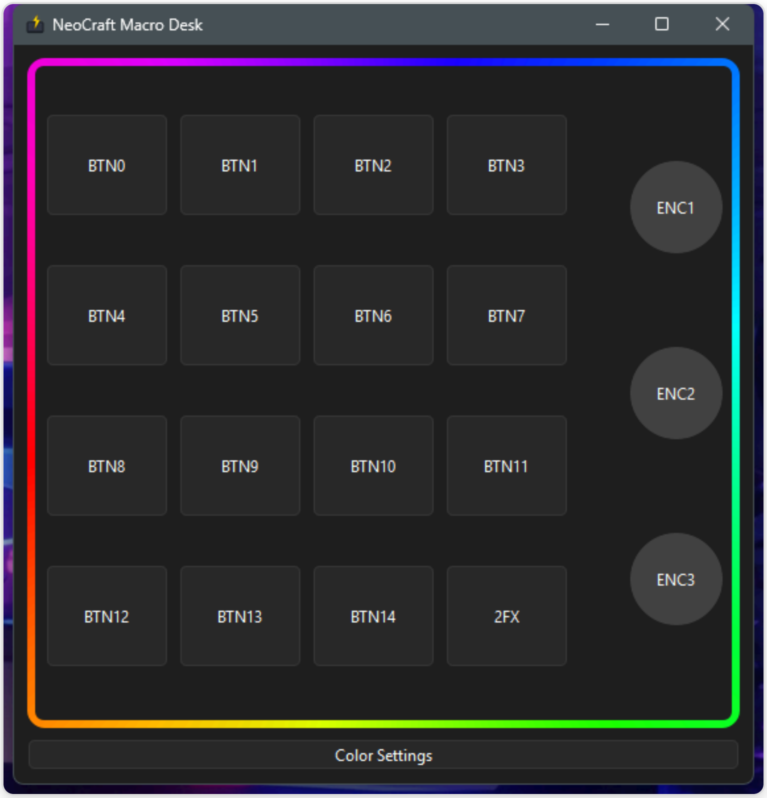
The app connects automatically — no setup dialog, no drivers to install separately (Windows' built-in USB-serial driver handles the Pro Micro). It finds the pad by its USB identity, not a fixed COM port number, so it connects correctly no matter which COM port Windows assigns it (which can change if you plug it into a different USB port).

If the LEDs stay solid red after launching the app: the app isn't seeing the pad at all — check the USB cable/connection, and confirm it shows up in Windows Device Manager under Ports (COM & LPT).

To uninstall: Start Menu → NeoCraft Macro Desk → "Uninstall NeoCraft Macro Desk," or Windows Settings → Apps. Your saved button/encoder configuration is kept (in case you reinstall later) — see Where your settings are saved if you want to remove it manually too.

Part 3 — How to use each function

The main window

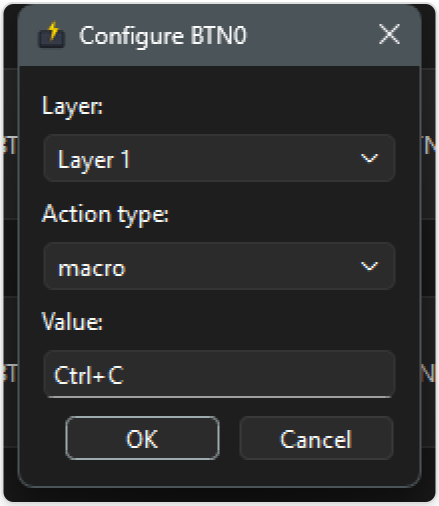


Note: this screenshot predates a small layout change — the single bottom button is now three side by side: Color Settings, Settings, and Help (see below).

A 4×4 grid of buttons plus the 3 encoders on the right. The animated border shows the pad's current LED pattern live. Click **any button** to configure it. Below the grid, three buttons: - **Color Settings** — change the LED pattern (see [Changing the LED pattern]). - **Settings** — 2FX timeout and app language (see [Changing the app language]). - **Help** — app version and links to this manual and the GitHub repo (see [Help]).

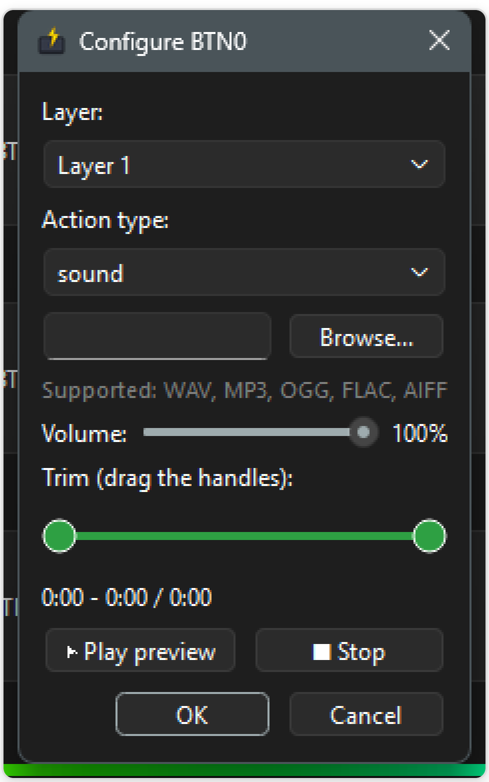
Assigning a keyboard shortcut or macro to a button

1. Click the button you want to configure (e.g. `BTN0`).
2. Choose the **Layer** (Layer 1 = normal, Layer 2 (2FX) = second function).
3. Set **Action type** to **Keyboard** or **Macro**.
4. **Keyboard**: type the combo directly, e.g. `ctrl+c`.
5. **Macro**: click into the field and press the actual key combo on your keyboard — it's captured live.
6. Click **OK** to save.



Assigning a sound to a button

1. Click the button, set **Action type** to **Sound**.
2. Click **Browse...** and pick an audio file (WAV, MP3, OGG, FLAC, or AIFF).
3. Adjust **Volume** with the slider (0–100%).
4. Drag the two handles under **Trim** to play only part of the clip — the label below shows the selected start/end time and the clip's total length.
5. Use ▶ **Play preview** / ■ **Stop** to check it sounds right before saving.
6. Click **OK**.



Assigning an OBS scene switch

1. Click the button, set **Action type** to **OBS Scene**.

2. Type the exact scene name as it appears in OBS.
3. Click **OK**.

This requires OBS Studio running with its WebSocket server enabled (OBS 28+ has this built in — Tools → WebSocket Server Settings).

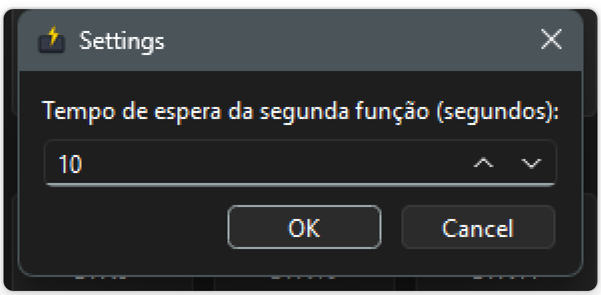
Clearing a button

Set its **Action type** to **Empty** and click **OK**.

Using the 2FX second layer

Tap the `2FX` key on the pad (bottom-right of the grid). The app border flashes red while armed. The next button you press runs its **Layer 2** action instead of Layer 1, then the pad returns to normal automatically. Tap `2FX` again before pressing anything else to cancel without running an action.

To change how long Layer 2 stays armed before auto-canceling, click the **Settings** button below the grid:



Changing the app language

Open the same **Settings** dialog and use the **Language** / **Idioma** dropdown at the bottom — pick **English** or **Português** and click **OK**. The change applies immediately: any dialog you open next (button config, encoder config, color settings) shows the new language right away, no restart needed. The **Color Settings/Settings/Help** buttons and the tray menu's Open/Quit labels update immediately too.

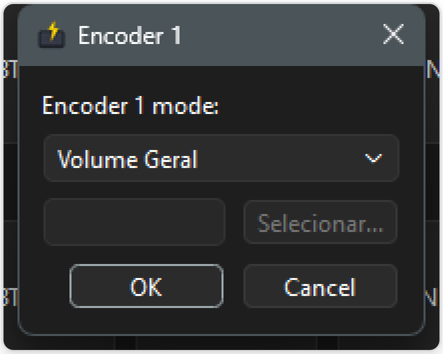
Help

Click the **Help** button below the grid for the app's version number and two links — one to this manual on GitHub, one to the project's GitHub repo. Both open in your default browser.

Configuring an encoder

1. Click an encoder (`ENC1` , `ENC2` , or `ENC3`) in the app window.
2. Choose the mode:

3. **System Volume** (**Volume Geral** in Português) — controls your system's overall Windows volume.
4. **Application** (**Aplicativo**) — controls one specific application's volume independently (click **Select...** and pick its `.exe`). This works correctly even for apps that run as many processes at once, like Chromium-based browsers.
5. Click **OK**.



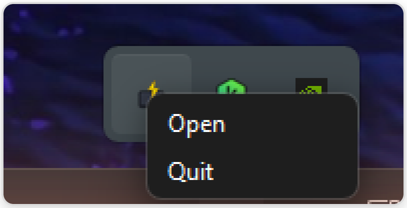
Once configured: **turn** the encoder to adjust volume up/down, **click** it (push down) to mute/unmute.

Changing the LED pattern

Click **Color Settings** at the bottom of the main window.

- **Pattern**: Solid Color, Breathing, Rainbow Wave, or Color Cycle.
- **Color**: pick a color on the wheel — only used by Solid Color and Breathing (grayed out otherwise, since Rainbow Wave and Color Cycle generate their own colors).
- **Brightness** / **Speed**: sliders, applied immediately and saved with the rest of your settings.

The system tray icon



Right-click (or on some systems, left-click) the tray icon for: - **Open** — bring the main window back. - **Quit** — fully close the app (button/encoder mappings stop working until you reopen it, and the pad's LEDs go solid red).

Where your settings are saved

Your button mappings, encoder targets, and color settings are stored in:

```
%APPDATA%\NeoCraft Macro Desk\config.json
```

This file persists across app updates and reinstalls. Deleting it resets everything to defaults the next time the app starts.